Shivarama Krishna Koyalakonda

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About

Data Scientist and Cloud Engineer with 3+ years of experience in data science, machine learning, data analytics, and AWS. Completed 40+ projects across domains, delivering scalable and data-driven solutions using modern ML and cloud tools. Skilled in building scalable data pipelines, developing predictive models, and delivering actionable business insights. Passionate about leveraging AI and automation to simplify workflows, enhance monitoring, and drive customer delight

Experience

Missouri S&T — June 2025 – Present
Volunteer Research Assistant — Rolla, MO, USA

- Conducted content analysis on 4–5 scholarly papers, extracting recurring codes and themes to define key variables for ML-driven social science research.
- Performed systematic content analysis on the CDP 2019 dataset (10K+ company responses), curating labeled data for supervised ML model development.
- Utilized Python NLP libraries to preprocess unstructured text, increasing annotation efficiency by 25% and enabling consistent labeling across thousands of records.
- Explored automation of qualitative content analysis by experimenting with tools such as NVivo, Atlas.ti, and MAXQDA, aiming to reduce manual coding time by 40%.
- Developed pipelines to align annotated data with supervised ML objectives, improving feature extraction quality by 15% and reducing misclassification errors.

Missouri S&T — January 2025 – May 2025
Graduate Teaching Assistant (ML Algorithms & Applications) — Rolla, MO, USA

- Preprocessed and analyzed the USPTO patents dataset to identify 50+ key variables, enabling classification of companies engaged in AI/ML innovation.
- Applied advanced feature engineering techniques, reducing data noise by 30% and improving downstream ML classification accuracy by 18%.
- Leveraged Google Cloud Storage and BigQuery to process and manage large-scale patent data that exceeded Excel's capacity, improving processing efficiency by 45%.
- Utilized Google Cloud services for scalable storage, querying, and data preprocessing, ensuring smooth handling of multi-million-row datasets and reducing computation time by 40%.

Cognizant — March 2022 – September 2022
AWS Cloud Data Engineer — Hyderabad, TG, India

- Developed scalable data pipelines for processing 1M+ of records daily, a framework applicable to Medicaid/Medicare claims datasets.
- Developed 7+ data pipelines to prepare training data for machine learning models, processing data daily across AWS services including S3, Redshift, EMR, and Athena.
- Built and optimized feature engineering pipelines using PySpark and SQL, reducing data preparation time by 50% and accelerating overall model retraining cycles.
- Automated ML workflows by integrating SageMaker training jobs with ingestion pipelines, reducing manual effort by 70% and cutting retraining cycle time from 3 days to <1 day.
- Created dashboards and reports using AWS Athena and Power BI, providing actionable insights to business units and reducing manual reporting time by 30%

Verzeo — April 2021 – August 2021
Machine Learning Intern — India

- Developed predictive and classification models, achieving up to 92% accuracy on test datasets by leveraging advanced feature engineering and hyperparameter tuning,
- Built and trained GANs to generate synthetic datasets, reducing the need for manual data augmentation and enhancing training data diversity, leading to a 40% improvement in model performance and efficiency.
- Engineered a high-performance computer vision model using Fast R-CNN with ResNet-50 to detect visible aging signs on facial images, attaining over 90% accuracy and optimizing the system for real-time responsiveness on a dataset exceeding 10,000 samples.
- Collaborated with cross-functional teams, gathering 5+ project requirements and delivering data models and reports that

improved business decision-making efficiency by 25%.

Freelance March 2020 – March 2022
Freelance Machine learning & Data Science Engineer India

- Conducted data analysis on large datasets using Python and SQL, extracting key insights that improved model accuracy by 15% and supported data-driven decision-making.
- Developed predictive models using regression and classification techniques, achieving up to 92% accuracy, and presented findings through interactive dashboards in Power BI.

Projects

Enhancing Resolution of images using GANs

- Created a deep learning model using GANs to improve image resolution by 400% to support visual quality in consumer-facing applications while retaining realistic textures and details.
- Utilized a custom GAN architecture that decreased model training time by 30% by optimizing generator-discriminator cycles.

Object detection with RetinaNet

- Implemented RetinaNet on the COCO 2017 dataset, achieving a mAP of 37.8 at IoU thresholds 0.5:0.95, outperforming ResNet-50 (30.0 mAP) and VGG-16 (27.1 mAP) by 25–40%. This improvement is due to RetinaNet's effective handling of class imbalance through its Focal Loss function, enabling superior detection of small and rare objects.
- Enhanced object detection accuracy to support automation in real-time systems with 20% reduced latency, ensuring reliable performance in autonomous systems and industrial automation.

Ageing Signs detection with Fast R-CNN

- Developed an aging signs detection model using Fast R-CNN with a ResNet-50 backbone, achieving 92% detection accuracy across key facial features such as wrinkles, sagging, and pigmentation.
- Fine-tuned on a custom-labeled dataset of images, reducing inference time by 35%, enabling near real-time performance for potential deployment in skincare and health applications.

City bike data analysis in NYC using SAP

- Conducted a data analysis project using SAP Analytics Cloud to identify usage patterns and peak demand, enabling business units to optimize resource allocation and improve usage prediction accuracy by 20%.
- Developed automated KPI dashboards to streamline reporting, reducing manual reporting time by 50% and providing actionable insights for forecasting and strategic planning.

Skills

- **Software Engineering & Programming:** Python, Java, C++, OOP, System Design, REST APIs, Design Patterns, Git, GitHub, Unit Testing (PyTest, JUnit)
- **Algorithms & Data Structures :** Trees, Graphs, Heaps, Sorting, Recursion, Dynamic Programming, Problem Solving
- **Machine Learning & AI:** Machine Learning, Deep Learning, Feature Engineering, EDA, Model Evaluation, CNN, RNN, LSTM, GRU, GANs, Transfer Learning, BERT, GPT, Autoencoders
- **Data Science & Analytics:** HealthCare Analytics, Data Analysis, Statistical Testing, Hypothesis Testing, Regression Analysis, A/B Testing, Forecasting, KPI Tracking, Data Quality Validation
- **Data Visualization & BI Tools:** HealthCare Dashboards, Power BI (DAX, Data Modeling), Tableau, ThoughtSpot, SAP Analytics Cloud, Matplotlib, Seaborn
- **Cloud & MLOps:** AWS (S3, Redshift, EMR, Athena, SageMaker), Docker, Kubernetes, CI/CD, Jenkins, CloudWatch, Agile/DevOps Practices, GCP cloud services.
- **Data Engineering & Ware Housing:** ETL for healthcare claims data, SQL (PostgreSQL, MySQL, Snowflake, BigQuery), Denodo, ETL (Airflow, AWS Glue, dbt), Data Warehousing (Redshift, Snowflake, Azure Synapse)

Education

Missouri University of Science and Technology August 2023 – May 2025

Master of Science in Information Science and Technology Rolla, MO, USA

- **Relevant Coursework:** Use of Business Intelligence, Data Warehouse & Dashboards, Business Analytics & Data Science, Data Science & Machine Learning in Python

Osmania University May 2018 – May 2022

Bachelor of Technology in Computer Science Hyderabad, TG, India